PROBABILITY BASICS ASSIGNMENT

Question 1: A die is rolled. What is the probability of getting:
(a) An even number
(b) A number greater than 4

Answer:

When a die is rolled, the total possible outcomes are: {1, 2, 3, 4, 5, 6} > Total = 6

(a) Probability of getting an Even Number:

Even numbers on a die = {2, 4, 6} > Favorable outcomes = 3

P(Even) = favorable outcomes / Total outcomes
= 3 / 6
= 1/2

(b) Probability of getting a Number Greater than 4:

Numbers greater than 4 = {5, 6} > favorable outcomes = 2

P(Greater than 4) = 2 / 6
= 1/3

Question 2: In a class of 50 students — 20 like Mathematics (M), 15 like Science (S), 5 like both subjects. What is the probability that a student chosen at random likes Mathematics or Science?

Answer:

This question uses the Addition Rule of probability because the two events overlap (5 students like both).

Formula:
P(M or S) = P(M) + P(S) − P(M and S)

Step 1 - Find individual probabilities:

- P(M) = 20/50
- P(S) = 15/50
- P(M and S) = 5/50

Step 2 - Apply the formula:
P(M or S) = 20/50 + 15/50 − 5/50
= 30/50
= 3/5

Question 3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

Answer:

This is a Conditional Probability problem because the first event (drawing a red ball) has already occurred and affects the second draw.

Initial setup:

- Total balls = 3 red + 2 blue = 5 balls
- First ball drawn = Red (already happened)

After first red ball is removed:

- Remaining red balls = 3 − 1 = 2 red balls
- Remaining total balls = 5 − 1 = 4 balls

Probability that the second ball is also Red:

P(Second Red) = Remaining red balls / Remaining total balls
= 2 / 4
= 1/2

Question 4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

Answer:

The correct method to use is Stratified Sampling.

What is Stratified Sampling?
Stratified Sampling is a method where the population is first divided into distinct groups (called strata) based on a specific characteristic:- in this case, gender (boys and girls). Then a random sample is taken from each group separately.

Why NOT Simple Random Sampling?
Simple Random Sampling gives every individual an equal chance of being selected, but it does NOT guarantee proportional or equal representation of subgroups. In this school with 60% boys and 40% girls, a simple random sample of 100 students might give us 65 boys and 35 girls - or even 70 boys and 30 girls - purely by chance. Girls would be under-represented.

Why Stratified Sampling is the Right Choice:

- The population has two clearly defined groups: Boys (60%) and Girls (40%)
- We specifically want equal representation of both genders
- With Stratified Sampling, we can deliberately select, say, 50 boys and 50 girls to ensure both groups are equally represented
- This eliminates gender bias from the sample and makes the study results fair and accurate for both groups

Question 5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

Answer:

What is Sampling Error?
Sampling error is the difference between the value calculated from the sample and the true value of the entire population. It occurs because we study only a part of the population, not everyone.

Formula:
Sampling Error = Sample Mean − Population Mean

Given:

- Population Mean (μ) = 160 cm
- Sample Mean ($\bar{x}$) = 158 cm

Calculation:
Sampling Error = (158 – 160) cm
= −2 cm

Question 6: The population mean salary is ₹50,000 with σ = ₹5,000. If we take a sample of 100 employees, what is the Standard Error of the Mean (SEM)?

Answer:

What is Standard Error of the Mean (SEM)?
The Standard Error of the Mean measures how much the sample mean is expected to vary from the true population mean. It tells us the accuracy of our sample estimate. A smaller SEM means our sample mean is a more reliable estimate of the population mean.

Formula:
SEM = $\sigma / \sqrt{n}$

PROBABILITY BASICS ASSIGNMENT

Given:

- Population Standard Deviation (σ) = ₹5,000
- Sample Size (n) = 100

Calculation:
SEM = 5000 / √100
= 5000 / 10
= 500x`

Question 7: In a group of 100 students :

40 like Cricket (C)

30 like Football (F)

10 like both Cricket and Football.

Find the probability that a student likes at least one sport.

Answer:

"At least one sport" means the student likes Cricket or Football or both this is a classic Addition Rule problem.

Formula:
P(C or F) = P(C) + P(F) - P(C and F)

Step 1 -Find individual probabilities:

- P(C) = 40/100 = 0.40
- P(F) = 30/100 = 0.30
- P(C and F) = 10/100 = 0.10

Step 2 - Apply Addition Rule:
P(C or F) = 0.40 + 0.30 - 0.10
= 0.60

Clarification:
There is a 60% probability that a randomly chosen student from this group likes at least one sport - Cricket or Football or both.

The remaining 40% students like neither sport.

Question 8: From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

Answer:

This is a Multiplication Rule Without Replacement problem. Since the first card is not returned before drawing the second, the total cards and number of Aces both decrease after the first draw.

Total Aces in a deck = 4
Total cards = 52

Step 1 - Probability that the 1st card is an Ace:
P(1st Ace) = 4/52 = 1/13

Step 2 - Probability that the 2nd card is also an Ace (given 1st was an Ace):
After removing one Ace: Remaining Aces = 3, Remaining cards = 51

P(2nd Ace) = 3/51 = 1/17

Step 3 - Multiply both probabilities:
P(Both Aces) = P(1st Ace) × P(2nd Ace)
= 4/52 × 3/51
= 12/2652
= 1/221

Clarification:
The probability that both cards drawn are Aces is 1 in 221, which is approximately 0.45% a very rare event. This makes intuitive sense because there are only 4 Aces in a 52-card deck, so drawing two consecutively without replacement is quite unlikely.

Question 9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Answer:

Given:

- Defective rate = 2% = 0.02
- Non-defective rate = 1 − 0.02 = **0.98**
- Number of bulbs selected = 5
- Each bulb is independent of the others

PROBABILITY BASICS ASSIGNMENT

This uses the Multiplication Rule for Independent Events:

P(All 5 non-defective) = P(1st non-defective) * P(2nd non-defective) * ... * P(5th non-defective)
= 0.98 * 0.98 * 0.98 * 0.98 * 0.98
= **$(0.98)^5$**

Step by step:

- ⇨ 0.98 * 0.98 = 0.9604
- ⇨ 0.9604*0.98 = 0.941192
- ⇨ 0.941192*0.98 = 0.922368
- ⇨ 0.922368*0.98 = 0. 922368
- ⇨ 0. 922368*0.98 = 0.903920
- ⇨ 0.9039 or 90.39%

Clarification:
There is approximately a 90.39% probability that all 5 randomly chosen bulbs are non-defective.

This makes logical sense since only 2% of bulbs are defective, the chance of picking 5 good ones in a row is quite high. However, as you increase the number of bulbs tested, this probability will gradually decrease.

Question 10: Differentiate between Discrete and Continuous Random Variables with examples.

Answer:

What is a Random Variable?
A Random Variable is a variable that takes a numerical value based on the outcome of a random experiment. It converts uncertain outcomes into numbers so they can be mathematically analyzed.

Random Variables are of two types:

1. Discrete Random Variable

A Discrete Random Variable takes countable, specific values - usually whole numbers (integers). The values are separate and distinct with clear gaps between them. You can list all possible values.

Characteristics:

- Values are countable (finite or countably infinite)
- No fractional or decimal values between counts
- Arises from counting processes
- Uses Probability Mass Function (PMF)

Examples:

- Number of heads when tossing a coin 3 times > possible values: 0, 1, 2, 3
- Number of students absent in a class > 0, 1, 2, 3...
- Number of defective items in a batch of 100
- Number of goals scored in a football match
- Number of customers entering a shop per hour

2. Continuous Random Variable

A Continuous Random Variable can take any value within a given range or interval - including decimals and fractions. You cannot list all possible values because there are infinitely many of them between any two points.

Characteristics:

- Values are uncountable and infinite within a range
- Can take any decimal or fractional value
- Arises from measurement processes
- Uses Probability Density Function (PDF)

Examples:

- Height of a student > 165.3 cm, 172.8 cm, 158.45 cm
- Temperature on a given day > 36.6°C, 37.2°C
- Weight of a fruit > 2.45 kg, 1.87 kg
- Time taken to complete a task > 23.7 minutes, 45.2 minutes
- Monthly salary of an employee > ₹42,350.75

Conclusion:
The key difference is simple - if you count something, it is Discrete. If you measure something, it is Continuous. This distinction is critical in data analytics because it determines which statistical methods and probability distributions you apply to the data.